

CROSS-ANATOMY DATA FUSION AND COMPATIBILITY MODELING EMBODIMENTS

In various embodiments, measurements obtained from the intraluminal compliance and distance profiling device may be integrated with data obtained from additional sensing devices or user profiles to generate composite models describing interaction compatibility, alignment characteristics, or physiologic coordination between different anatomical systems.

The computational system may combine distance measurements, compliance characteristics, deformation response profiles, and stability metrics with complementary measurements acquired from external geometry scanners, longitudinal physiologic monitors, or wearable biometric devices. Data fusion may occur locally within a user device or within a remote computational platform.

In certain embodiments, cross-anatomy modeling may generate derived parameters describing relative alignment, proportional matching, deformation compatibility, comfort prediction, or interaction stability. Such derived metrics may be calculated using statistical correlation, rule-based modeling, machine learning analysis, or hybrid computational techniques.

The system may normalize measurements from different devices into standardized reference frameworks prior to fusion. Normalization may include dimensional scaling, confidence weighting, calibration correction, or conversion into abstracted indices designed to preserve privacy while enabling meaningful comparison.

In some implementations, compatibility modeling may operate independently of explicit anatomical labeling. Abstracted structural and compliance parameters may be compared mathematically without revealing sensitive anatomical information, thereby supporting privacy-preserving interaction analysis.

Data fusion workflows may also support longitudinal adaptation in which compatibility models improve over time as additional measurements are collected. Computational learning systems may refine prediction accuracy using repeated sessions, aggregated anonymized datasets, or population-level modeling approaches.

In certain embodiments, the system may generate compatibility indicators, alignment scores, or interaction profiles derived from combined datasets. Outputs may be expressed as numerical scores, graphical indicators, symbolic representations, or other non-explicit visual formats designed for intuitive interpretation.

The disclosed architecture further allows interoperability between independently owned devices or profiles. Secure matching protocols may permit comparison of derived parameters between consenting users while maintaining encryption, anonymization, and user-controlled authorization mechanisms.

Accordingly, cross-anatomy data fusion embodiments extend the functionality of the disclosed device beyond isolated measurement, enabling coordinated multi-device analysis, adaptive personalization, and scalable compatibility modeling within a broader physiologic computation ecosystem.